

Quasi-Poisson models

Last time: quasi-Poisson models

Poisson:

- + Mean = λ_i
- + Variance = λ_i

quasi-Poisson:

- + Mean = λ_i
 - + Variance = $\phi \lambda_i$
 - + Variance is a linear function of the mean
- ϕ : dispersion parameter

Last time: quasi-Poisson models

Poisson:

- + $\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim N(0, 1)$
- + Use $N(0, 1)$ for hypothesis testing and confidence intervals for one parameter
- + Comparing nested models (drop-in-deviance): χ^2 test

quasi-Poisson:

- + $SE(\hat{\beta}_j)_{QP} = \sqrt{\hat{\phi}} \times SE(\hat{\beta}_j)_{Pois}$
- + $\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)_{QP}} \sim t_{n-p}$
- + Use t_{n-p} for hypothesis testing and confidence intervals for one parameter
- + Comparing nested models: F test

Class activity

Work on the class activity (handout).

Class activity

Which model do you think is more appropriate for the data?

Model 2 (Quasi-Poisson)

• $\hat{\phi} = 5.52$ (variance is $> 5 \times \text{mean}$)

• GOF test:

H_0 : model 1 is a good fit

H_A : model 1 is not a good fit

test statistic: residual dev = 11540

df = 2004

p-value ≈ 0
 $\text{pchisq}(11540, \text{df} = 2004, \text{lower.tail} = F)$

$$H_0: \beta_3 = 0$$

$$H_A: \beta_3 \neq 0$$

Class activity

Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with no college education, after accounting for sex, age, diabetes, and BMI?

$$t = \frac{0.0165 - 0}{0.0126 \sqrt{5.52}} = 0.557 < 1.303$$

Model 1:

Estimate Std. Error z value Pr(>|z|)

...

education2 0.0164626 0.0126312 1.303 0.192462

...

$$p\text{-value} \approx 0.58 > 0.192$$

Model 2:

$$z_{\text{pvt}}(0.557, \text{df}=2004, \text{lower.tail}=F)$$

(Dispersion parameter for quasipoisson family taken to be 5.519388)

Nested tests with quasi-Poisson models

```
m2 <- glm(cigsPerDay ~ male + age + education + diabetes + BMI,  
          data = smokers, family = quasipoisson)  
summary(m2)
```

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.837192	0.115099	24.650	< 2e-16	***
male	0.456170	0.026095	17.481	< 2e-16	***
age	-0.006806	0.001595	-4.267	2.07e-05	***
education2	0.016463	0.029675	0.555	0.579	
education3	0.016427	0.037706	0.436	0.663	
education4	-0.015033	0.040477	-0.371	0.710	
diabetes	-0.025398	0.092606	-0.274	0.784	
BMI	0.005001	0.003304	1.513	0.130	

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

$$H_0: \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A: \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

linear regression: $F = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}}) / (df_{\text{reduced}} - df_{\text{full}})}{SSE_{\text{full}} / df_{\text{full}}}$

Nested tests with quasi-Poisson models

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

- ✚ In Poisson regression, we would use a likelihood ratio test
- ✚ However, the quasi-Poisson model includes the estimated dispersion, $\hat{\phi}$. We need to use an **F-test** instead = # parameters tested

$$F = \frac{(\text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}) / (df_{\text{reduced}} - df_{\text{full}})}{\hat{\phi}_{\text{full}}}$$

$$\hat{\phi}_{\text{full}} = \frac{1}{n-p} \sum_{i=1}^n e_{(mi)}^2$$

q = # parameters tested

Under H_0 , $F \sim F_{q, n-p}$ p = # parameters in full model

numerator df $\rightarrow$ q denominator df $\rightarrow$ $n-p$

Nested tests with quasi-Poisson models

```
m1 <- glm(cigsPerDay ~ male + age + education +  
          diabetes + BMI,
```

← full model

```
          data = smokers, family = quasipoisson)
```

```
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,  
          data = smokers, family = quasipoisson)
```

← reduced model

`m1$deviance`

```
## [1] 11539.56
```

`m2$deviance`

$$F = \frac{(11543.8 - 11539.56) / (2007 - 2004)}{S.S.Z}$$

```
## [1] 11543.84
```

`m1$df.residual`

$$\text{under } H_0, F \sim F_{3, 2004}$$

```
## [1] 2004
```

`m2$df.residual`

```
## [1] 2007
```

Nested tests with quasi-Poisson models

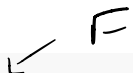
```
m1$deviance
```

```
## [1] 11539.56
```

```
m2$deviance
```

```
## [1] 11543.84
```

```
pf(0.258, 3, 2004, lower.tail=F)
```



```
## [1] 0.8556648
```

Limitations of quasi-Poisson models

Important: quasi-Poisson models do *not* describe a full distribution for the response (there is no "quasi-Poisson" distribution). They describe *only* the mean and variance of the response.

As a result, one cannot calculate probabilities for a quasi-Poisson:

```
m2 <- glm(cigsPerDay ~ male + age + diabetes + BMI,  
          data = smokers, family = quasipoisson)  
rootogram(m2)
```

Error in prodist.glm(object) :
family is not associated with a full probability distribution

Class activity

Work on the class activity (handout).